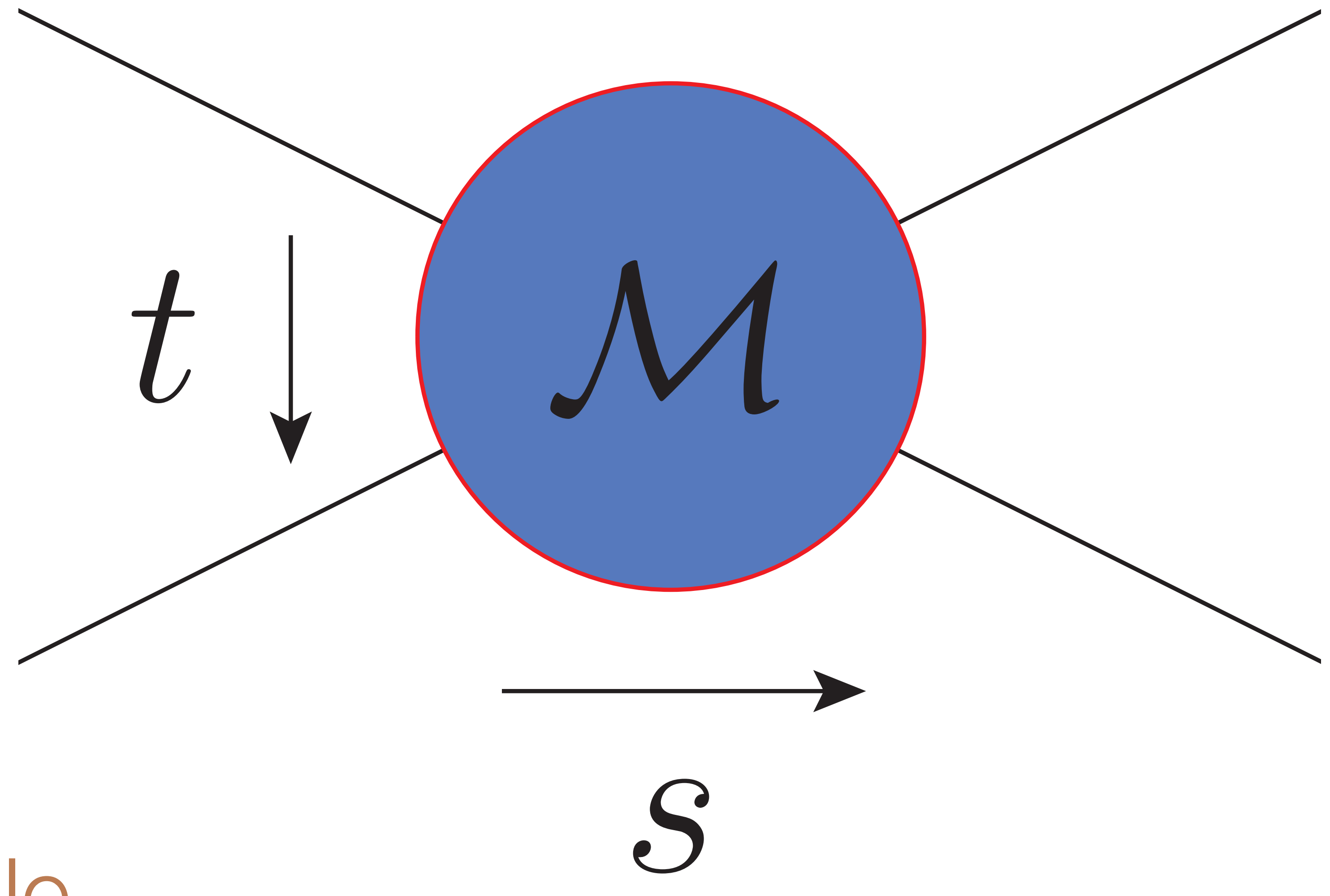
The High-Energy Limit of 2->2 Scattering Amplitudes at NNLL

- Calum Milloy, Università di Torino and INFN Torino

[Einan Gardi, Giulio Falcioni, CM, Leonardo Vernazza, 2012.00613; + Niamh Maher 2105.XXXXX]



$s \gg t$

Leading logarithms resummed via famous Regge pole

$$\mathcal{M}^{\mathrm{LL}}_{ij\to ij}(s,t) = \left(\frac{s}{-t}\right)^{C_A\,\alpha_g(t)} \mathcal{M}^{\mathrm{tree}}_{ij\to ij}$$

Regge trajectory

$$\alpha_g = \left(\frac{\alpha_s}{\pi}\right)\frac{1}{2\epsilon} + \mathcal{O}(\alpha_s^2)$$

$$\mathcal{M}^{\mathrm{tree}}_{ij\to ij} = g_s^2\,\frac{2s}{t}\,\mathbf{T}_i\cdot\mathbf{T}_j$$

colour charge of parton i, j
we will keep expressions representation independent

Motivation

- Understand all-order structure of amplitudes
- Provide constraints for infrared divergences
- Towards an interacting Reggeon field theory

How to systematically deal with subleading logarithms?

Consider symmetry properties of amplitude

$$\mathcal{M}^{(\pm)}(s,t) = \frac{1}{2}\Big[\mathcal{M}(s,t) \pm \mathcal{M}(u,t)\Big]$$

+/- $\longleftrightarrow$ even/odd amplitude

By defining symmetric-even logarithm the odd/even amplitude corresponds to the real/imaginary part

$$L = \frac{1}{2}\left(\log\frac{-s-i0}{-t} + \log\frac{-u-i0}{-t}\right)$$

Formulating highly energetic partons as Wilson lines

[Caron-Huot '13; Caron-Huot, Gardi, Vernazza '17]

$$U(z_\perp) = \mathbf{P}\exp\left[ig_s \mathbf{T}^a \int_{-\infty}^{\infty} dx^+ A_+^a(x^+, x^- = 0, z_\perp)\right]$$

rapidity regulator

$$\eta = \frac{1}{2}\log\frac{p_+}{p_-}$$

$$T^a_{i,L} = [\mathbf{T}^a U(z_i)]\,\frac{\delta}{\delta U(z_i)}$$

$$T^a_{i,R} = [U(z_i)\mathbf{T}^a]\,\frac{\delta}{\delta U(z_i)}$$

our parton is a collection of such Wilson lines

evolves according to Balitsky-JIMWLK

$$\frac{d}{d\eta}\left|\psi_i\right\rangle = -H\left|\psi_i\right\rangle \qquad H = \frac{\alpha_s}{2\pi^2}\int d^d z_i d^d z_j d^d z_0 \frac{z_{0i}\cdot z_{0j}}{(z_{0i}^2 z_{0j}^2)^{1-\epsilon}}\left\{T^a_{i,L}T^a_{j,L} + T^a_{i,R}T^a_{j,R} - U^{ab}_{\mathrm{ad}}(z_0)\left(T^a_{i,L}T^b_{j,R} + T^a_{j,L}T^b_{i,R}\right)\right\}$$

Target Projectile

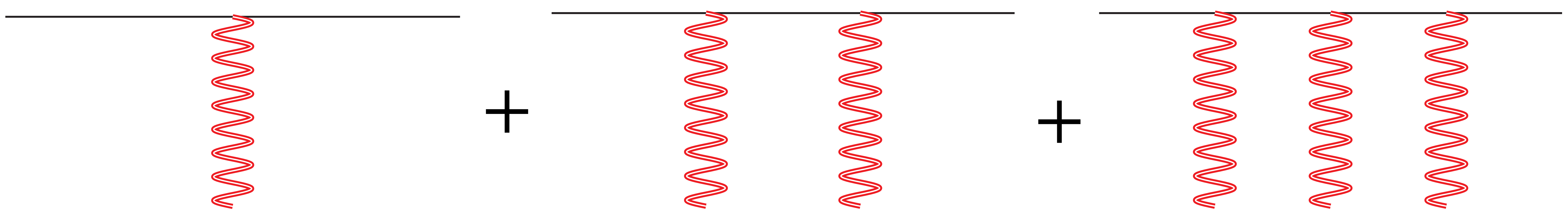
The amplitude is then written as $\mathcal{M}_{ij\to ij} \sim \langle\psi_j|e^{-HL}|\psi_i\rangle$

Evolve so that they are at equal rapidity

Expand Wilson line in *Reggeons*

Reggeon field

$$U = \exp\left[ig_s \mathbf{T}^a W^a\right] \sim$$



Only odd/even number of Reggeons contribute to the odd/even amplitude

In this framework the NLL divergences have been resummed and finite parts known to very high loop order

[Caron-Huot, Gardi, Reichel, Vernazza '17, '20]

$$\hat{\mathcal{M}}^{(+)}_{\mathrm{NLL}}\Big|_s = \frac{i\pi}{L(C_A - \mathbf{T}_t^2)}\left(1 - R(\epsilon)\frac{C_A}{C_A - \mathbf{T}_t^2}\right)^{-1} \times \left[\exp\left\{\frac{B_0(\epsilon)}{2\epsilon}\frac{\alpha_s}{\pi}L(C_A - \mathbf{T}_t^2)\right\} - 1\right]\mathbf{T}^2_{s-u}\mathcal{M}^{(\mathrm{tree})} + \mathcal{O}(\epsilon^0).$$

Can we achieve the same at NNLL?

The odd NNLL amplitude

$$\mathcal{M}_{ij\to ij} \sim \langle \psi_j | e^{-HL} | \psi_i \rangle$$

Expand in terms of the coupling and logarithms

$$|\psi_i\rangle = \sum_{k=1}^{\infty} g_s^k \, |\psi_{i,k}\rangle \qquad \hat{H}_{m\to n} \sim g_s^{|m-n|}$$

Expand Hamiltonian to describe transitions between different Reggeon (W) states

We get the all-order expression for the n-loop amplitude

$$\hat{\mathcal{M}}^{(-),\mathrm{NNLL}}_{ij\to ij} \sim \left(\frac{\alpha_s}{\pi}\right)^n \Big(\langle\psi_{j,3}|\, \hat{H}_{3\to 3} + \langle\psi_{j,1}|\, \hat{H}_{3\to 1} \Big)$$

$$\times\, \hat{H}^{n-4}_{3\to 3}$$

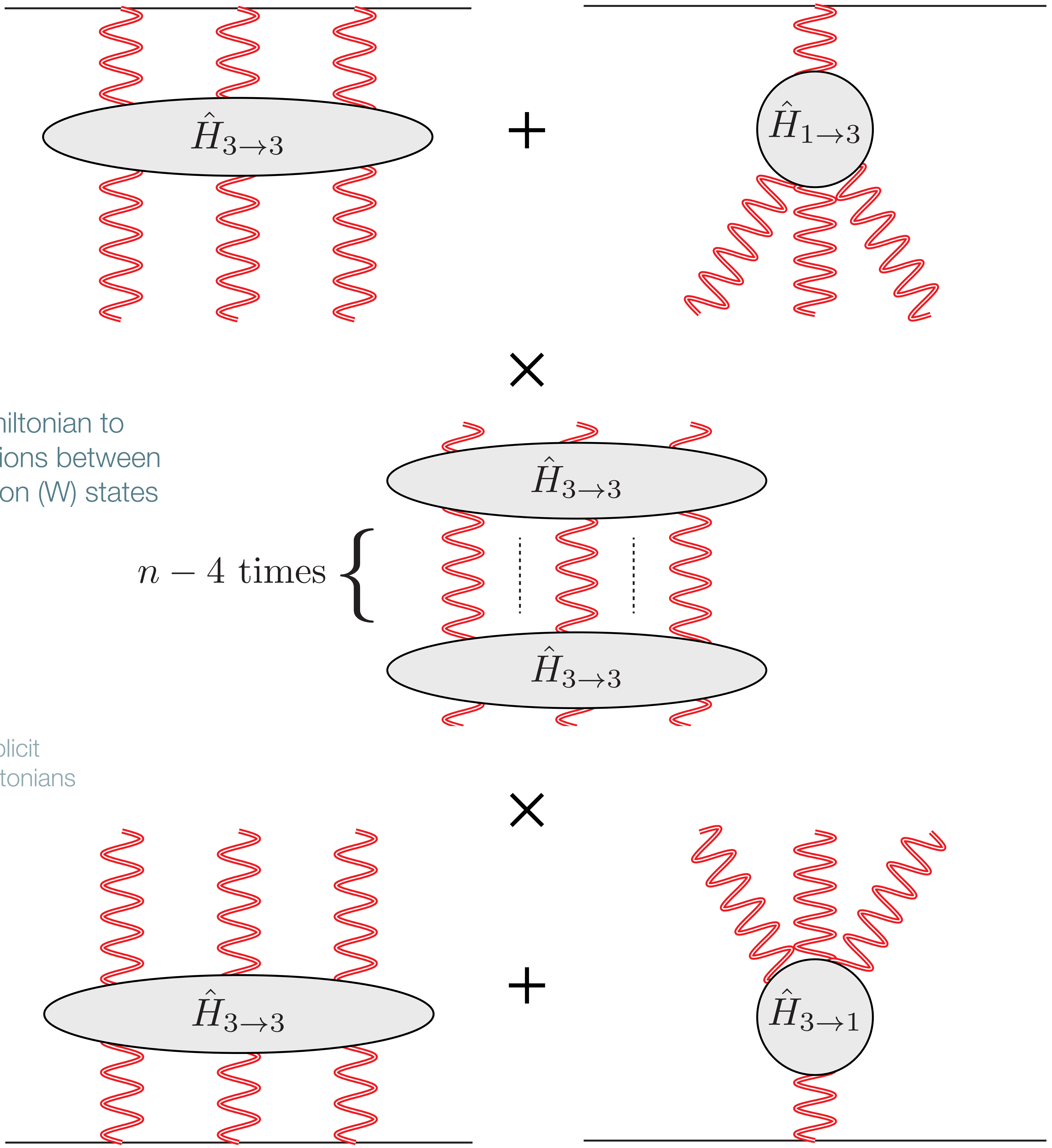
$$\times \Big(\hat{H}_{3\to 3}\, |\psi_{j,3}\rangle + \hat{H}_{1\to 3}\, |\psi_{j,1}\rangle \Big)$$

$$+\,\text{explicit two-loop information}$$

reduced amplitude, see slide 7

see slide 8 for the explicit momentum-space Hamiltonians

Only require Leading Order Balitsky-JIMWLK Hamiltonian and results from two-loop amplitudes (impact factors)



The four-loop NNLL amplitude

Integrals are four-loop propagator-type

$$f_\epsilon = \zeta_3 + \frac{3}{2}\epsilon\zeta_4$$

We can write the colour structure in terms of

$$\mathbf{T}_s = \mathbf{T}_1 + \mathbf{T}_2 \qquad \mathbf{T}_t = \mathbf{T}_1 + \mathbf{T}_4 \qquad \mathbf{T}_u = \mathbf{T}_1 + \mathbf{T}_3$$

from which we get the quadratic operators

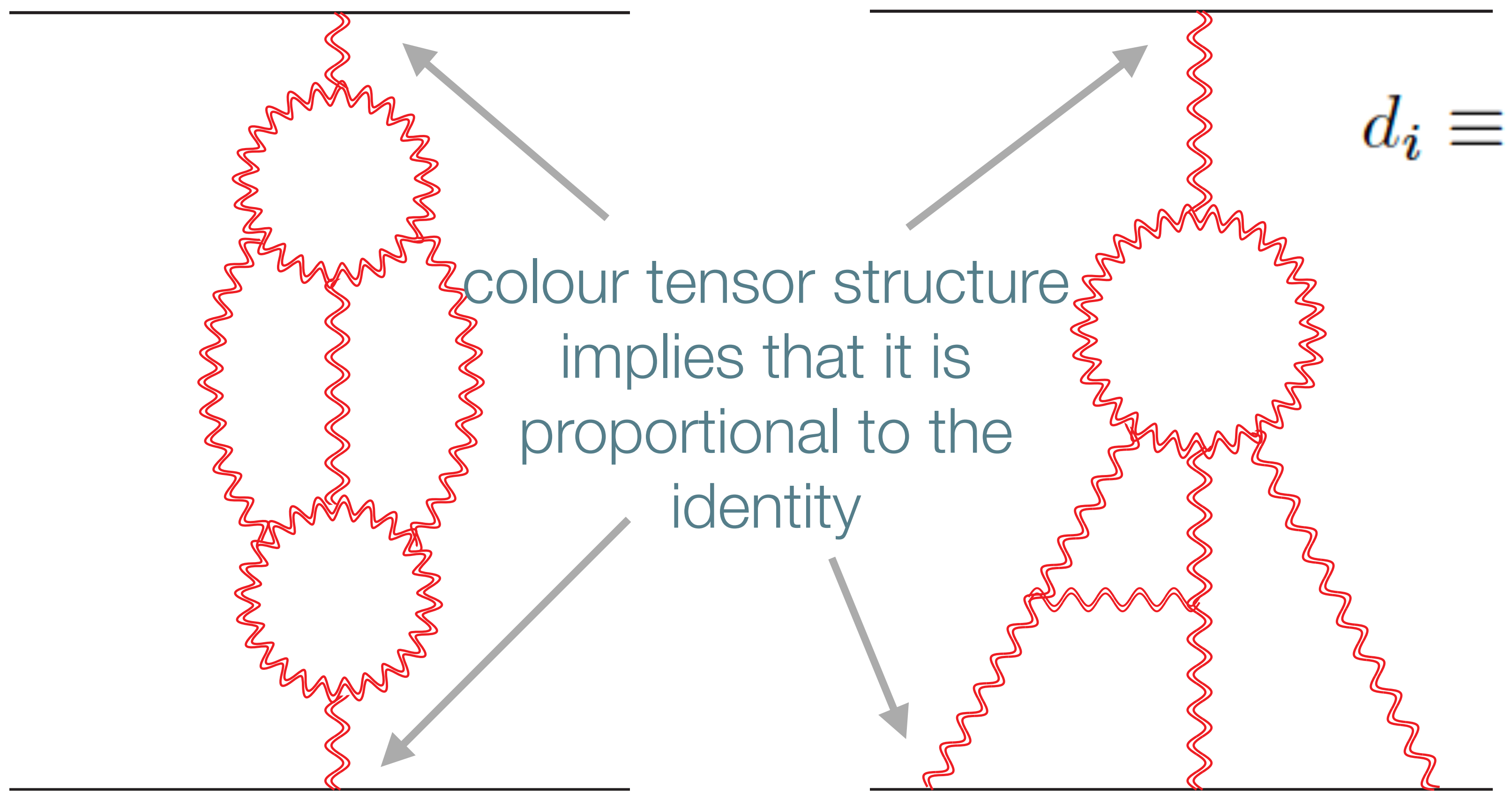
$$\mathbf{T}^2_{s-u} = \frac{1}{2}(\mathbf{T}^2_s - \mathbf{T}^2_u) \qquad \mathbf{T}^2_t \mathcal{M}^{\rm tree}_{ij\to ij} = C_A \mathcal{M}^{\rm tree}_{ij\to ij}$$

for instance one structure is for more details on colour, see slide 9

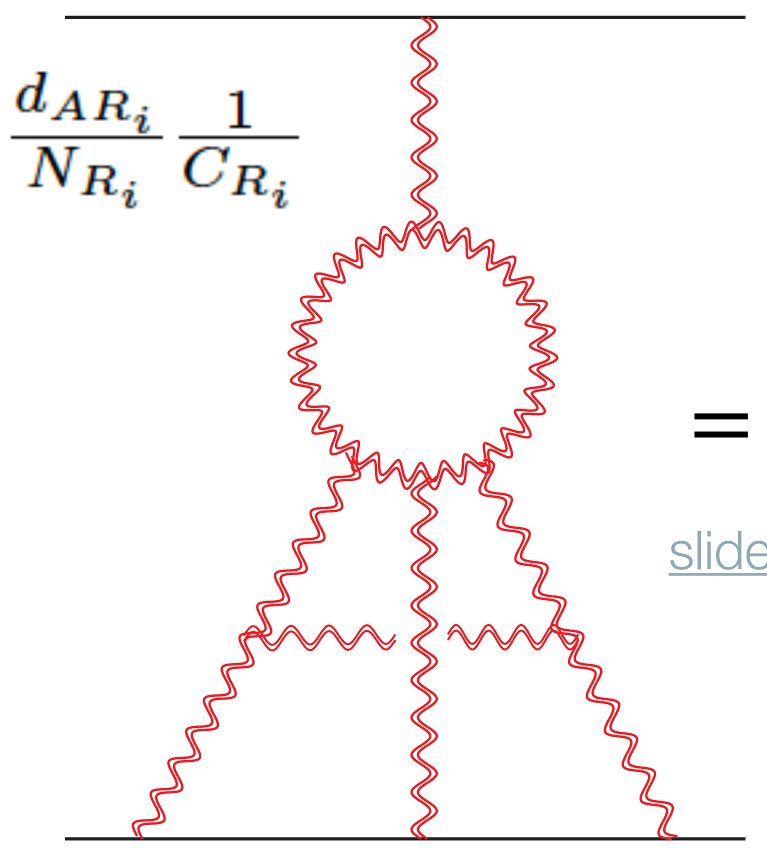
$$\mathbf{C}^{(-4)}_{\mathcal{M}} = -\frac{2}{3}\mathbf{T}^2_{s-u}(\mathbf{T}^2_t)^2\mathbf{T}^2_{s-u} - \frac{4}{3}C_A^2(\mathbf{T}^2_{s-u})^2 + \frac{4}{3}\mathbf{T}^2_t\mathbf{T}^2_{s-u}\mathbf{T}^2_t\mathbf{T}^2_{s-u} - \frac{16}{9}(\mathbf{T}^2_t)^2(\mathbf{T}^2_{s-u})^2 + \frac{22}{9}C_A\mathbf{T}^2_t(\mathbf{T}^2_{s-u})^2$$

and the full amplitude:

$$\hat{\mathcal{M}}^{(-),4,\rm NNLL}_{ij\to ij} = \frac{\pi^2}{144}\left[\mathbf{C}^{(-4)}_{\mathcal{M}}\frac{1}{\epsilon^4} + \mathbf{C}^{(-1)}_{\mathcal{M}}\frac{f_\epsilon}{\epsilon} + \mathcal{O}(\epsilon)\right]\mathcal{M}^{\rm tree}_{ij\to ij}$$



$$d_i \equiv \frac{d_{AR_i}}{N_{R_i}}\frac{1}{C_{R_i}}$$

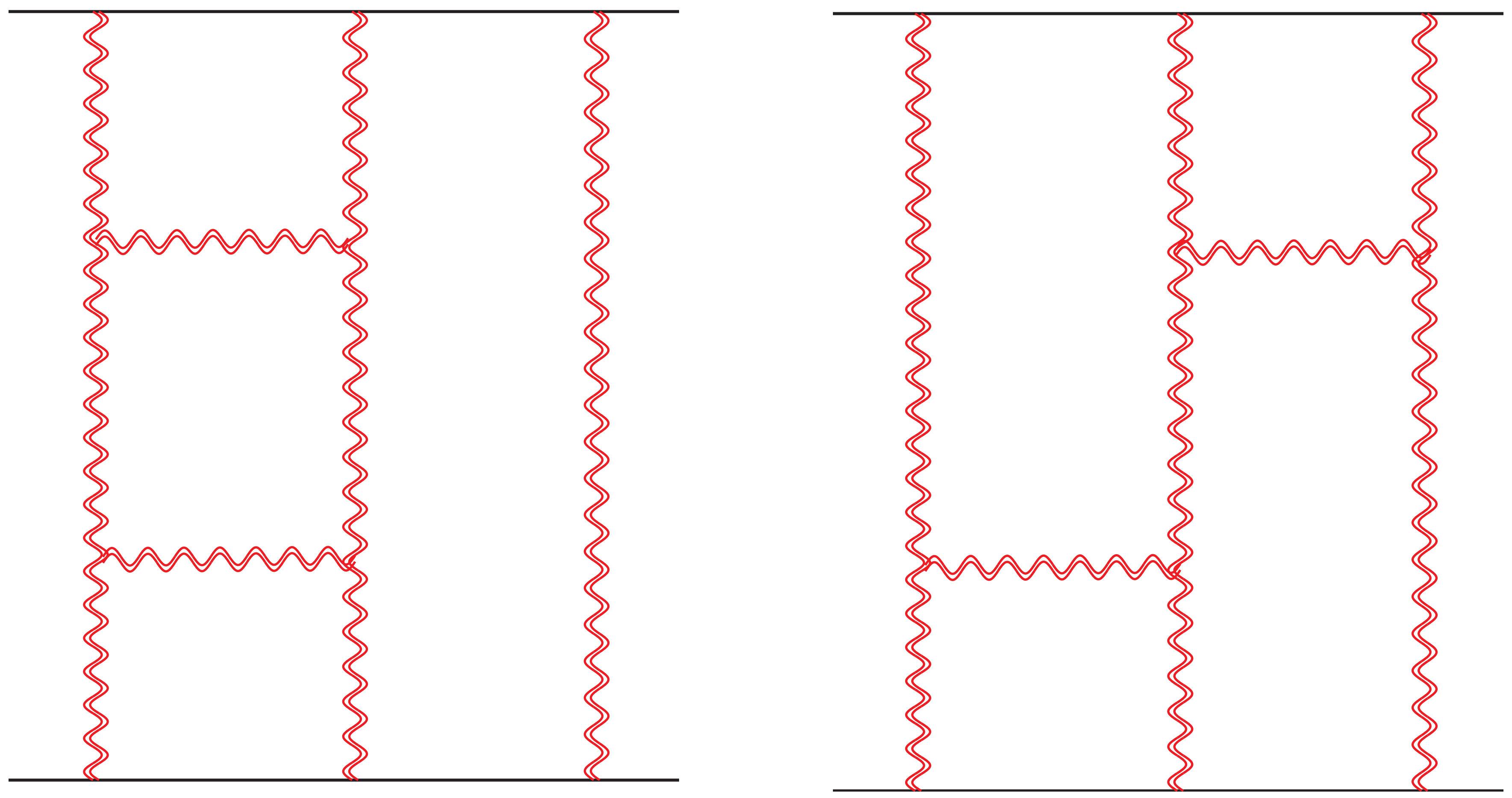


$$= 0$$

slide 10

$$\langle\psi_{j,1}|\hat{H}_{3\to1}\hat{H}_{1\to3}|\psi_{i,1}\rangle = \frac{1}{432}\left[-\left(\frac{C_A^2}{12} + \frac{d_{AA}}{N_A}\right)\frac{1}{\epsilon^4} + \left(\frac{101}{6}C_A^4 + 220\frac{d_{AA}}{N_A}\right)\frac{f_\epsilon}{\epsilon} + \mathcal{O}(\epsilon)\right]\langle\psi_{j,1}|\psi_{i,1}\rangle$$

$$\langle\psi_{j,1}|\hat{H}_{3\to1}\hat{H}_{3\to3}|\psi_{i,3}\rangle = \frac{C_A d_i}{144}\left[\frac{1}{\epsilon^4} - 208\frac{f_\epsilon}{\epsilon} + \mathcal{O}(\epsilon)\right]\langle\psi_{j,1}|\psi_{i,1}\rangle$$



$$\langle\psi_{j,3}|\hat{H}^2_{3\to3}|\psi_{i,3}\rangle = \frac{1}{144}\left[\frac{\mathbf{C}^{(4,-4)}_{33}}{\epsilon^4} + \frac{2f_\epsilon}{\epsilon}\mathbf{C}^{(4,-1)}_{33} + \mathcal{O}(\epsilon)\right]\langle\psi_{j,1}|\psi_{i,1}\rangle$$

Infrared Factorisation Comparison

we found	demand finite

$$\mathcal{M} = \mathbf{Z}\cdot\mathcal{H}, \qquad \mathbf{Z} = \mathcal{P}\exp\left\{-\frac{1}{2}\int_0^{\mu^2}\frac{d\lambda^2}{\lambda^2}\mathbf{\Gamma}\right\},$$

to find

Dipole formula in HE

$$\mathbf{\Gamma} = \frac{\gamma_K}{2}\left[L\mathbf{T}_t^2 + i\pi\mathbf{T}_{s-u}^2\right] + \Gamma_i + \Gamma_j + \mathbf{\Delta},$$

Explicit results in HE, in any gauge theory

NLL 4-loop $\mathbf{\Delta}^{(4,3)} = -i\pi\frac{\zeta_3}{24}[\mathbf{T}_t^2,[\mathbf{T}_t^2,\mathbf{T}_{s-u}^2]]\mathbf{T}_t^2$

NNLL 3-loop $\mathbf{\Delta}^{(3,1)} = i\pi\frac{\zeta_3}{4}[\mathbf{T}_t^2,[\mathbf{T}_t^2,\mathbf{T}_{s-u}^2]]$

$$\mathrm{Re}\left[\mathbf{\Delta}^{(4,2)}\right] = \zeta_2\zeta_3\,\mathbf{C}_{\mathbf{\Delta}}^{(4,2)}$$

$$\mathbf{C}_{\mathbf{\Delta}}^{(4,2)} = \frac{1}{4}\mathbf{T}_t^2[\mathbf{T}_t^2,(\mathbf{T}_{s-u}^2)^2] + \frac{3}{4}[\mathbf{T}_{s-u}^2,\mathbf{T}_t^2]\mathbf{T}_t^2\mathbf{T}_{s-u}^2 + \frac{d_{AA}}{N_A} - \frac{C_A^4}{24}.$$

Also the NNLL 4-loop hard function in N=4 sYM

$$\mathrm{Re}\left[\mathcal{H}_{\mathcal{N}=4}^{(4,2)}\right] = \left[\frac{C_A^4}{128}\zeta_3^2 + \frac{3}{16}\zeta_4\zeta_2\,\mathbf{C}_{\mathbf{\Delta}}^{(4,2)}\right]\mathcal{M}^{\mathrm{tree}}$$

Soft anomalous dimension 4-loop ansatz

$$\begin{aligned}\mathbf{\Delta} =& f(\alpha_s)\sum_{(i,j,k)}\mathcal{T}_{iijk} + \sum_{(i,j,k,l)}\mathcal{T}_{ijkl}F(\beta_{ijlk},\beta_{iklj};\alpha_s)\\ &+\sum_R g^R(\alpha_s)\left[\sum_{i<j}2\left(\mathcal{D}_{iijj}^R + 2\mathcal{D}_{iiij}^R\right)\log\frac{\mu^2}{-s_{ij}} + \sum_{(i,j,k)}\mathcal{D}_{ijkk}^R\log\frac{\mu^2}{-s_{ij}}\right]\\ &+\sum_R\sum_{(i,j,k,l)}D_{ijkl}^R G^R(\beta_{ijlk},\beta_{iklj};\alpha_s) + \sum_{(i,j,k,l)}\mathcal{T}_{ijkli}H_1(\beta_{ijlk},\beta_{iklj};\alpha_s)\end{aligned}$$

[Becher, Neubert '19]
slide 11

We act the ansatz on the tree-level

$$\mathbf{\Delta}\mathcal{M}_{\mathrm{tree}} \to (\text{function of } \mathbf{T}_{s-u}^2 \text{ and } \mathbf{T}_t^2)\mathcal{M}_{\mathrm{tree}}$$

Giving asymptotic limits of the four-loop functions

Combinations giving odd/even parts

$$F_{AS}^{(4)} \sim -\frac{i\pi}{24}\zeta_3 L^3 + \ldots$$

terms subleading in L

$$F_S^{(4)} \sim -\frac{1}{8}\zeta_2\zeta_3 L^2 + \ldots \qquad G^{(4),A} \sim \frac{1}{2}\zeta_2\zeta_3 L^2 + \ldots$$

$$H_{1AS}^{(4)} \sim 0\times L^3 + \ldots \qquad H_{1S}^{(4)} \sim 0\times L^2 + \ldots$$

Consistent with Vladimirov '17 observation that there is no colour odd structure

Extra material

Reduced amplitude

We find the more natural object to compute is the *reduced* amplitude

removes Regge trajectory

$$\hat{\mathcal{M}}_{ij\to ij} = (Z_i Z_j)^{-1} e^{-\mathbf{T}_t^2 \alpha_g(t) L} \mathcal{M}_{ij\to ij}$$

removes collinear divergences from legs

The expression for the amplitude is then modified to

$$\frac{i}{2s}\hat{\mathcal{M}}_{ij\to ij} \equiv \langle\psi_j|e^{-(H-H_{1\to 1})L}|\psi_i\rangle = \langle\psi_j|e^{-\hat{H}X}|\psi_i\rangle$$

where we define the Regge trajectory to be 1 to 1 transitions,
see other definitions motivated by Glauber SCET in [Rothstein, Stewart '16; Ridgway @ SCET 2020 & REF 2020]

At NNLL there is an ambiguity in how to define the Regge trajectory

Explicit momentum-space Hamiltonians

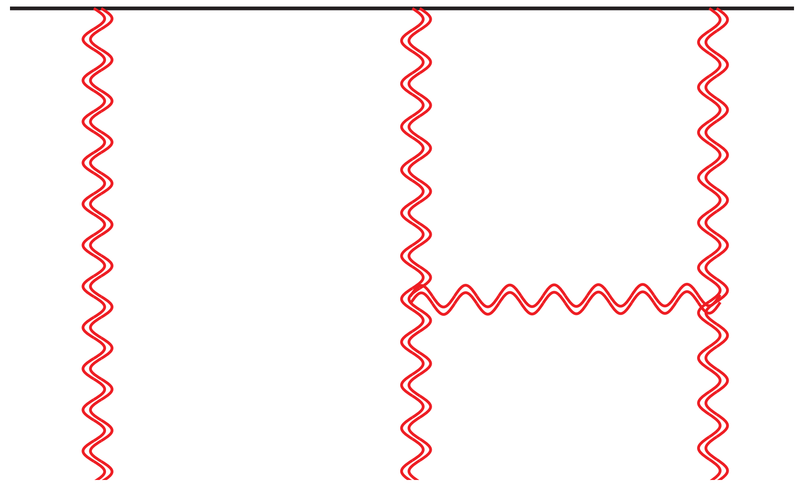
The diagonal transitions are

dresses one Reggeon with the trajectory

$$H_{k\to k} = -\int d^d p\, C_A \alpha_g(p) W^a(p) \frac{\delta}{\delta W^a(p)}$$

$$+\,\alpha_s \int d^d q d^d p_1 d^d p_2\, H_{22}(q;p_1,p_2) W^x(p_1+q) W^y(p_2-q) (F^x F^y)^{ab} \frac{\delta}{\delta W^a(p_1)} \frac{\delta}{\delta W^b(p_1)}$$

adds a *rung* between two Reggeons



Source of the difficulty at NNLL. Three Reggeons spoil the symmetry between colour and kinematics, which is there for two Reggeons (NLL).

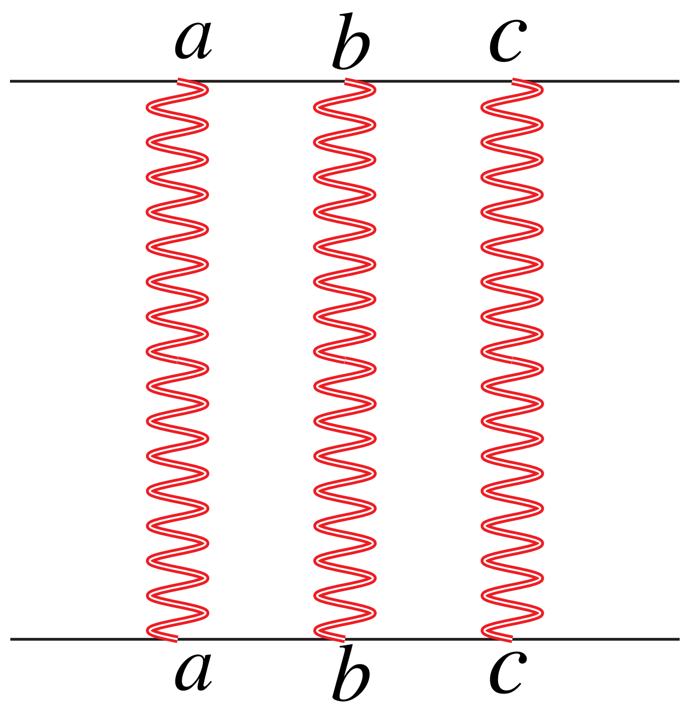
with kernel $H_{22}(q;p_1,p_2) = \dfrac{(p_1+p_2)^2}{p_1^2 p_2^2} - \dfrac{(p_1+q)^2}{p_1^2 q^2} - \dfrac{(p_2-q)^2}{q^2 p_2^2}$

The off-diagonal transitions are

$$H_{1\to 3} = \alpha_s^2 \int d^d p_1 d^d p_2 d^d p\ \mathrm{tr}\left[F^a F^b F^c F^d\right] W^b(p_1) W^c(p_2) W^d(p_3) H_{13}(p_1,p_2,p_3) \frac{\delta}{\delta W^a(p)}$$

with kernel $H_{13}(p_1,p_2,p_3) = \dfrac{r_\Gamma}{3\epsilon}\left[\left(\dfrac{\mu^2}{(p_1+p_2+p_3)^2}\right)^\epsilon + \left(\dfrac{\mu^2}{p_2^2}\right)^\epsilon - \left(\dfrac{\mu^2}{(p_1+p_2)^2}\right)^\epsilon - \left(\dfrac{\mu^2}{(p_2+p_3)^2}\right)^\epsilon\right]$

Simplifying Colour

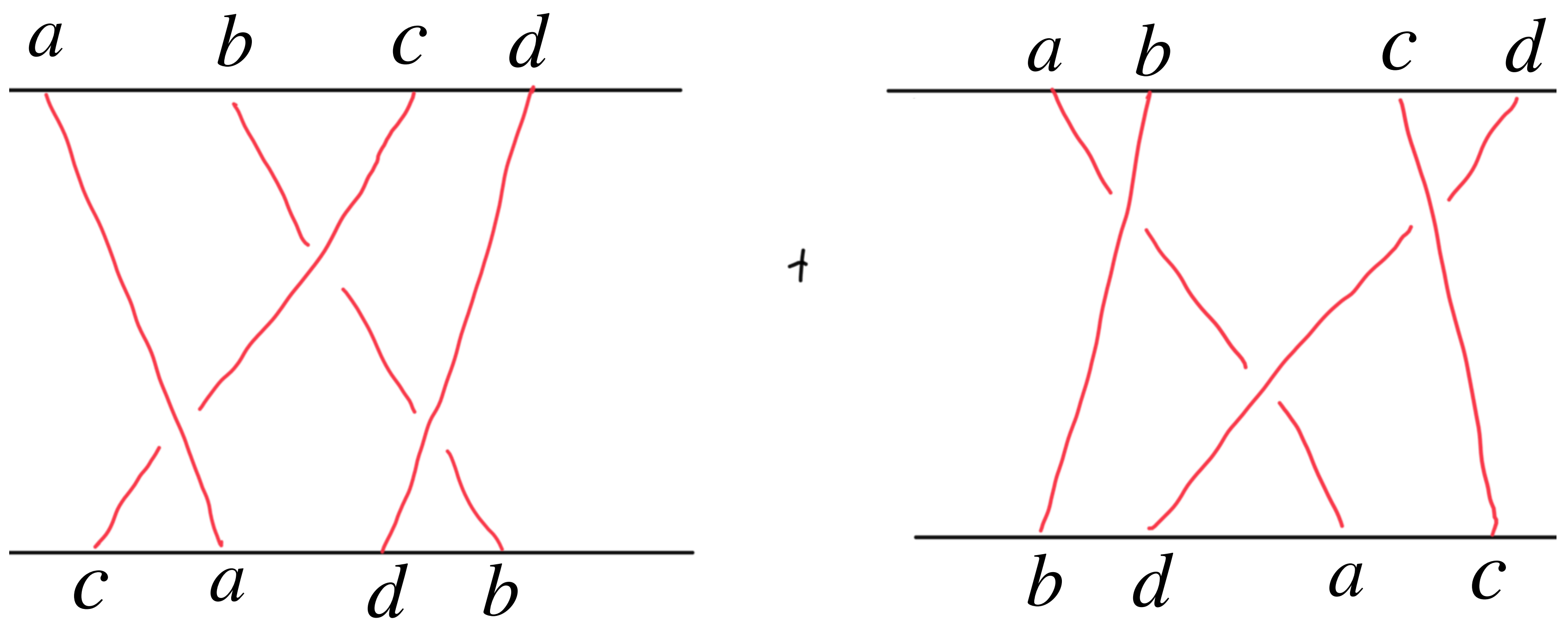


Simple example

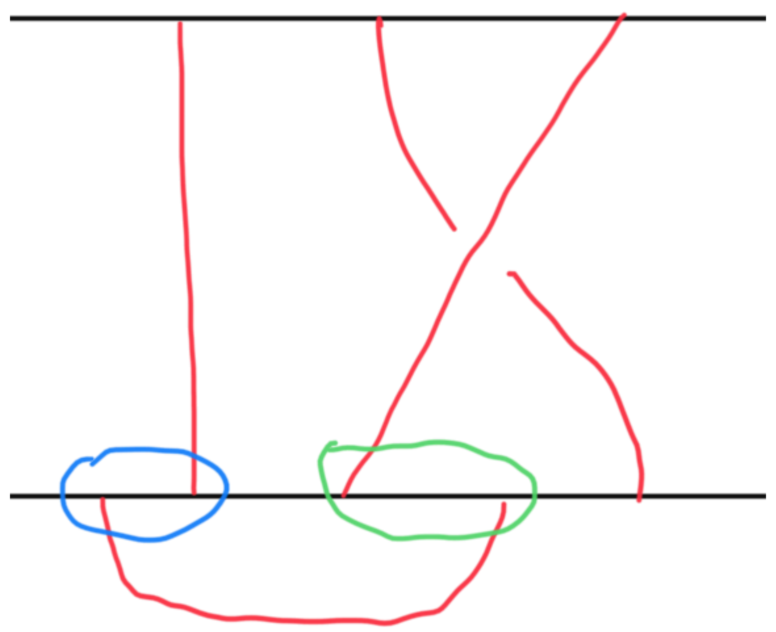
$$\mathbf{T}_i^a\mathbf{T}_i^b\mathbf{T}_i^c\mathbf{T}_j^a\mathbf{T}_j^b\mathbf{T}_j^c = (\mathbf{T}_i\cdot\mathbf{T}_j)(\mathbf{T}_i\cdot\mathbf{T}_j)(\mathbf{T}_i\cdot\mathbf{T}_j) = \frac{1}{2}\left(\mathbf{T}_{s-u}^2 - \frac{1}{2}\mathbf{T}_t^2\right)\frac{1}{2}\left(\mathbf{T}_{s-u}^2 - \frac{1}{2}\mathbf{T}_t^2\right)\mathcal{M}^{\mathrm{tree}}$$

It is natural to write in terms of a colour operator acting on the tree-level

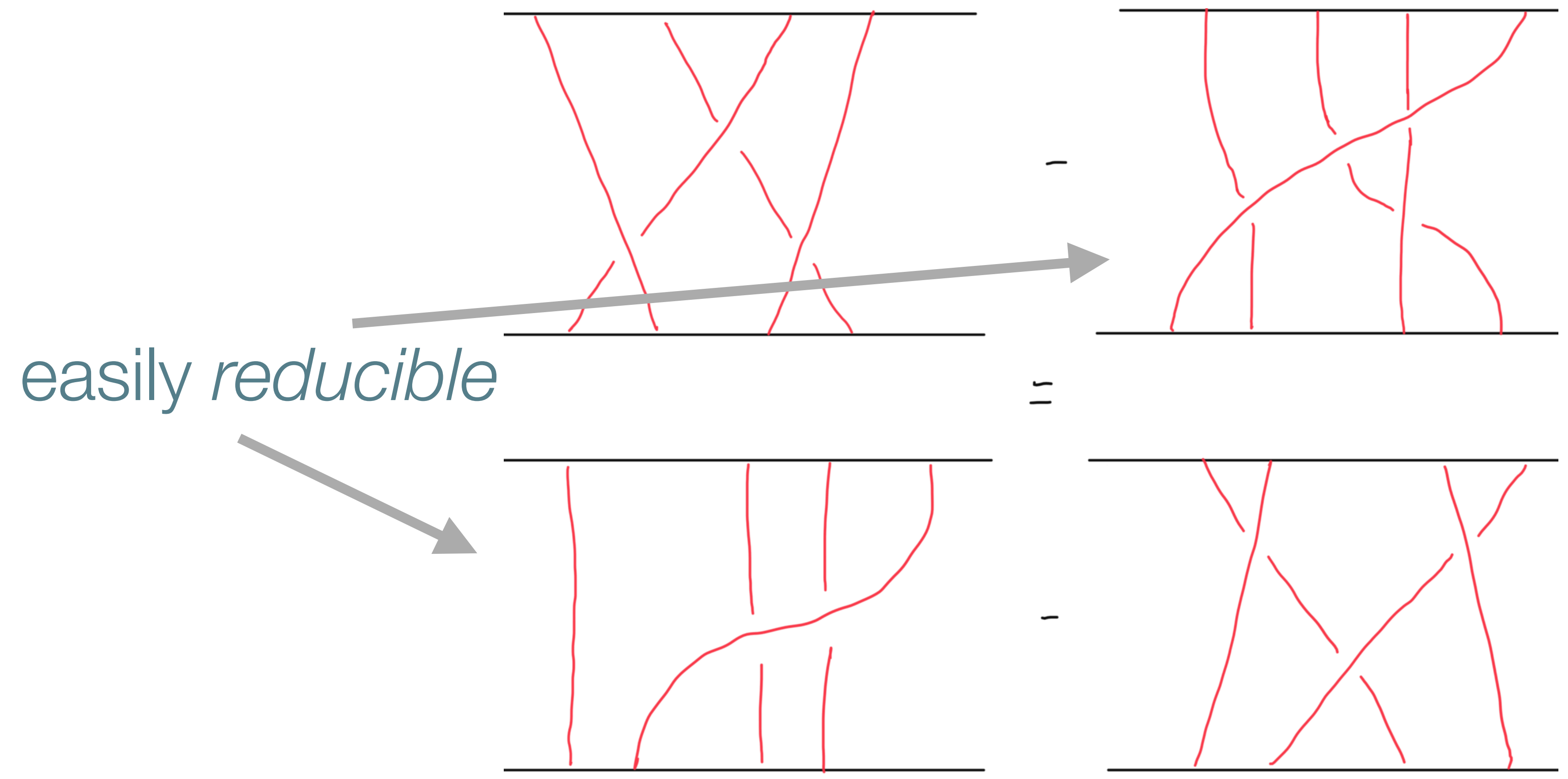
For some, we need to derive some non-trivial identities



not *reducible* to $\{\mathbf{T}_t^2, \mathbf{T}_{s-u}^2\}$ by standard techniques



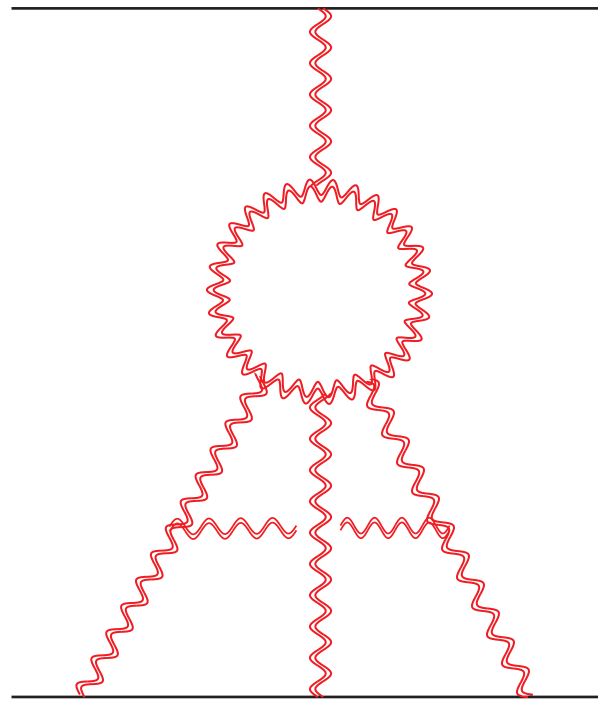
Consider the *boomerang graph*
Use commutation relation to swap the gluons attached in blue or green circles



Top/bottom line from green/blue circle
Thus we can relate the original sum (1 + 4) to reducible graphs (2 + 3)

There are also other relations involving five Reggeons that are required

Simplifying Colour



$$\sum_{\sigma\in\mathcal{S}_3}(F^xF^y)^{ab}\mathrm{tr}\left(F^xF^cF^yF^d\right)\mathbf{T}_j^d\mathbf{T}_i^{\sigma(a)}\mathbf{T}_i^{\sigma(b)}\mathbf{T}_i^{\sigma(c)}$$

It can be shown that this colour structure vanishes, independent of the Lie algebra

It can also be written in terms of the colour operators

$$\left(3C_A^2(\mathbf{T}_{s-u}^2)^2-2C_A\mathbf{T}_t^2(\mathbf{T}_{s-u}^2)^2-3C_A\mathbf{T}_{s-u}^2\mathbf{T}_t^2\mathbf{T}_{s-u}^2-(\mathbf{T}_t^2)^2(\mathbf{T}_{s-u}^2)^2+3\mathbf{T}_t^2\mathbf{T}_{s-u}^2\mathbf{T}_t^2\mathbf{T}_{s-u}^2\right)\mathbf{T}_i\cdot\mathbf{T}_j$$

We have an identity relating them when acting on the tree-level

However, sometimes it is impossible to write in terms of $\{\mathbf{T}_t^2,\mathbf{T}_{s-u}^2\}$

For instance, part of the N3LL three-loop term of the SAD

[Almelid, Duhr, Gardi '16; Caron-Huot, Gardi, Vernazza '17]

$$f^{abe}f^{cde}\Big[\{\mathbf{T}_t^a,\mathbf{T}_t^d\}\left(\{\mathbf{T}_{s-u}^b,\mathbf{T}_{s-u}^c\}+\{\mathbf{T}_{s+u}^b,\mathbf{T}_{s+u}^c\}\right)+\{\mathbf{T}_{s-u}^a,\mathbf{T}_{s-u}^d\}\{\mathbf{T}_{s+u}^b,\mathbf{T}_{s+u}^c\}\Big]-\frac{5}{8}C_A^2\mathbf{T}_t^2$$

but when we act only on the tree-level we do get

$$\left[\frac{7}{2}(\mathbf{T}_{s-u}^2)^2\mathbf{T}_t^2-\frac{5}{2}\mathbf{T}_{s-u}^2\mathbf{T}_t^2\mathbf{T}_{s-u}^2+\frac{1}{2}\mathbf{T}_t^2(\mathbf{T}_{s-u})^2-\frac{3}{8}C_A^3\right]\mathcal{M}^{\mathrm{tree}}$$

Up to four loops, at NNLL everything can be written in terms of $\{\mathbf{T}_t^2,\mathbf{T}_{s-u}^2\}$

Not true beyond since e.g. at N3LL we need acting on $\mathbf{T}_{s-u}^2\mathcal{M}^{\mathrm{tree}}$

The colour structures of the 4-loop SAD

[Becher, Neubert '19]

$$\mathcal{M} = \mathbf{Z} \cdot \mathcal{H}, \qquad \mathbf{Z} = \mathcal{P} \exp\left\{-\frac{1}{2}\int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2}\boldsymbol{\Gamma}\right\},$$

cusp anomalous dimension

collinear anomalous dimension

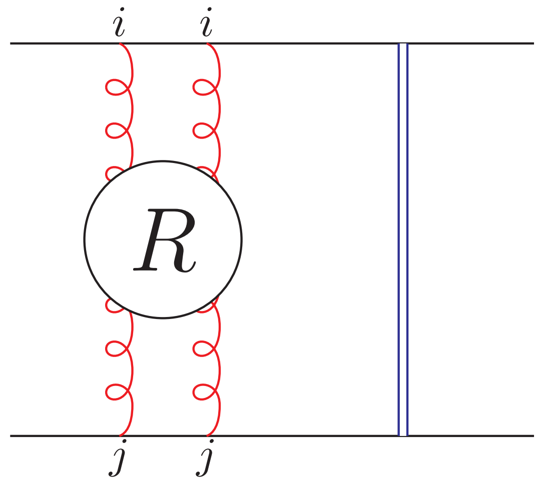
$$\boldsymbol{\Gamma} = \frac{\gamma_K}{2} \sum_{(i,j)} \mathbf{T}_i \cdot \mathbf{T}_j \log \frac{\mu^2}{-s_{ij}} + \sum_i \gamma_i + \boldsymbol{\Delta}$$

$$\begin{aligned}\boldsymbol{\Delta} =& f(\alpha_s) \sum_{(i,j,k)} \mathcal{T}_{iijk} + \sum_{(i,j,k,l)} \mathcal{T}_{ijkl} F(\beta_{ijlk}, \beta_{iklj}; \alpha_s) \\ &+ \sum_R g^R(\alpha_s) \left[\sum_{i<j} 2\left(\mathcal{D}^R_{iijj} + 2\mathcal{D}^R_{iiij}\right) \log \frac{\mu^2}{-s_{ij}} + \sum_{(i,j,k)} \mathcal{D}^R_{ijkk} \log \frac{\mu^2}{-s_{ij}} \right] \\ &+ \sum_R \sum_{(i,j,k,l)} D^R_{ijkl} G^R(\beta_{ijlk}, \beta_{iklj}; \alpha_s) + \sum_{(i,j,k,l)} \mathcal{T}_{ijkli} H_1(\beta_{ijlk}, \beta_{iklj}; \alpha_s)\end{aligned}$$

$$\begin{aligned}\mathcal{T}_{ijkl} &= f^{ade} f^{bce} \left(\mathbf{T}_i^a \mathbf{T}_j^b \mathbf{T}_k^c \mathbf{T}_l^d\right)_+ \\ \mathcal{D}^R_{ijkl} &= d_R^{abcd} \mathbf{T}_i^a \mathbf{T}_j^b \mathbf{T}_k^c \mathbf{T}_l^d \\ \mathcal{T}_{ijklm} &= i f^{adf} f^{bcg} f^{efg} \left(\mathbf{T}_i^a \mathbf{T}_j^b \mathbf{T}_k^c \mathbf{T}_l^d \mathbf{T}_m^e\right)_+\end{aligned}$$

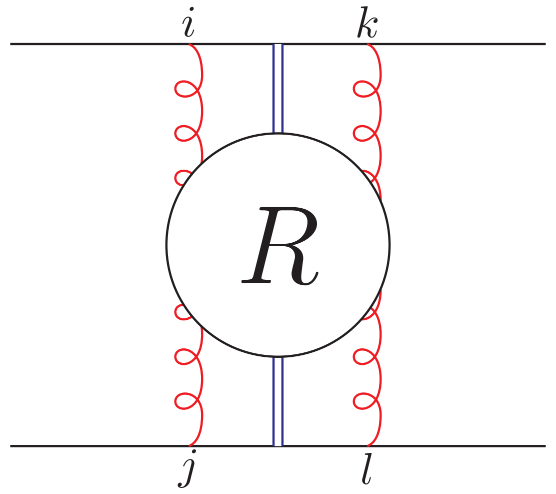
Example diagrams

$$\mathcal{D}^R_{iijj} \mathbf{T}_i \cdot \mathbf{T}_j \sim$$



the blue line is the tree-level

$$\mathcal{D}^R_{ijkl} \mathbf{T}_i \cdot \mathbf{T}_j \sim$$



partons in representation R run in the loop

$$\mathcal{T}_{ijkli} \mathbf{T}_i \cdot \mathbf{T}_j \sim$$